ChatGPT

Executive Summary

This report provides a comprehensive blueprint for preparing a manuscript for submission to *Gastroenterology*. It covers the required **manuscript structure** (title page, abstract, introduction, methods, results, discussion, acknowledgments, references, figure legends, tables) and includes sample text templates for each section. All recommendations strictly follow *Gastroenterology*'s author guidelines [1] [2], ICMJE authorship criteria [3], and applicable reporting standards (CONSORT, STROBE, PRISMA, etc.) [4] [5]. Key submission requirements are summarized in a detailed **checklist**: word and abstract length, format of references, limits on figures/tables, supplementary policy, conflict-of-interest and ethics statements, data sharing, and trial registration. Example references are formatted in the journal's citation style. Suggested **figures and tables** (with captions and mock data) are outlined. Finally, a proposed **timeline and task breakdown** (with a Gantt chart) guides the writing process from planning to submission. This report does not assume a specific topic or data; instead it provides flexible guidance (with notes where study type or data would vary).

Manuscript Structure and Outline

Manuscripts submitted as full **Original Articles** to *Gastroenterology* should include the following sections in order, with content as specified by the journal's author instructions [1] [6]:

- **Title Page:** Title; all authors' names and institutional affiliations; full contact details for the corresponding author; statements of authorship contribution, conflict of interest, and data/material availability [1]. Corporate/group authorship is allowed with special listing [7]. (See *Template Section* below.)
- **Structured Abstract:** ≤260 words with four subheadings – *Background and Aims*, *Methods*, *Results*, *Conclusion* – plus 4–5 keywords [6]. The abstract should succinctly state the research question, study design, primary findings, and major conclusion (omitting references) [6].
- **"What You Need to Know" Box:** A boxed summary on page 2 with five headings: **Background and Context**, **New Findings**, **Limitations**, **Clinical Research Relevance**, and **Basic Research Relevance**. Provide ~25–30 words under the first three headings and ~50–60 words under the last two [8]. Use non-technical language to state the study's importance and limits.
- **Lay Summary:** A one-sentence, ~25–30 word plain-language summary of the main findings (for TOC/social media) [9]. Write in simple terms without jargon.
- **Introduction:** Brief background (pathophysiology, epidemiology, or unmet need) and rationale, ending with a clear statement of the study's aim or hypothesis. Cover relevant literature and gap in knowledge.
- **Materials and Methods:** Immediately after the Introduction [10]. Describe study design, setting, subjects, interventions, and analytical methods. Include details on enrollment, inclusion/exclusion criteria, interventions, and measurements. State statistical methods and software (detail suffices for reproducibility [11]). Cite relevant guidelines (CONSORT, STROBE, ARRIVE, etc.) and include any checklists (see *Submission Checklist*). For human studies, include IRB/ethics approval and informed consent; for animal studies, state animal care approval (ARRIVE standards) [12] [13].
- **Results:** Summarize key findings in logical order. Report participant flow (e.g., number screened, excluded, analyzed) and baseline characteristics (often Table 1). Present primary outcomes with effect sizes, confidence intervals, and exact *p*-values (avoid reporting "NS" or p<0.05 alone). Show main results in tables/figures with legends. Use the journal's preferred statistical format (e.g.

mean±SD or median [IQR]). Highlight any subgroup or sensitivity analyses. All numerical data cited in text must match tables/figures.

- **Discussion:** Interpret results in context. Start with a summary of main findings. Compare to prior studies and mechanistic expectations. Discuss strengths and limitations (including bias/uncertainty). Explain clinical relevance and basic science implications. Conclude with overall interpretation and future directions. Do not overstate conclusions.
- **Acknowledgments:** Credit funding sources, grant numbers, technical assistance, and individuals who contributed but do not meet ICMJE authorship criteria [3]. Disclose roles of sponsors.
- **References:** Numbered list in the journal's style (first author et al., year; no article title) [14]. Use *Gastroenterology* style: e.g. "Smith AB, et al. Gastroenterology 2026;170:123–130." List shared-first-author names in **bold** and include the standard bold-author note [15]. Cite data sets using the format "[dataset] Author. Title. Repository. Year. DOI." as required [16].
- **Figure Legends and Tables:** Provide captions for all figures/fig panels and tables. Each figure or table should be cited in the text in order. Tables should be self-explanatory (with notes/abbreviations defined in legend). Limit to 7 combined tables+figures (no more than 7 in total, see *Checklist*). Number figures and tables sequentially.

The above structure strictly follows *Gastroenterology*'s format [2] [15]. Generic headings (e.g. "Background") should not be used; instead use the terms specified above.

Section Templates (Examples)

Below are illustrative template sentences and paragraphs for each section. These should be adapted to the specific study type (clinical trial, cohort, lab experiment, etc.) and data. Unspecified study details are noted where content would vary.

- **Title Page (Example):**
- *Title:* "Impact of [Intervention] on [Outcome] in [Population/Model]" (concise and specific).
- *Authors:* e.g. "Jane Doe^1, John Smith^2*, et al." (with shared-first authors in bold and † footnote as needed).
- *Affiliations:* e.g. "^1Dept. of Gastroenterology, University X; ^2Dept. of Epidemiology, University Y."
- *Corresponding Author:* Name, address, email.
- *Author Contributions:* Brief statement (e.g. "J.D. designed the study, conducted experiments, and wrote the manuscript; J.S. performed data analysis; both authors approved the final version").
- *Conflict of Interest:* "No conflicts to disclose." Or list relevant disclosures.
- *Data/Materials Availability:* e.g. "Deidentified data and analytic code available at [link] [17]." Or "All data included in article."
- *Funding:* e.g. "Supported by NIH grant XYZ."
- *Data Transparency Statement (for clinical trials):* e.g. "Individual participant data available at [repository] 3–5 years post-publication" [17].

- *Ethics Statement:* e.g. "IRB approval obtained from University X (protocol #1234); participants gave written informed consent." [12]

- **Abstract (Example):** *Background & Aims:* "Chronic [Disease] affects [population] and may be influenced by [factor]. We investigated whether [intervention/exposure] alters [biological measure/outcome] in [model/population]." *Methods:* "In this [study design] we enrolled N [patients/mice] with [criteria]. We measured [outcome] using [technique]. Statistical analysis included [t-test, regression, etc.]." *Results:* "[Summarize main quantitative findings: e.g.

"Treatment significantly increased [outcome] (mean±SD: X vs Y, p<0.01)."] *Conclusion:* "This study shows that [intervention] [improves/affects] [outcome] in [population], suggesting [implication]." *(Ensure total ≤260 words.)* Keywords: GI, [Key concepts]. [6]

- **Introduction (Example):** "[Disease/Mechanism] is a leading cause of [morbidity] worldwide. [Brief epidemiology or background, e.g. "Recent studies have identified factor X as a mediator of disease progression"]. However, it remains unclear whether [critical question]. We hypothesized that [intervention or exposure] would [expected effect] on [outcome] in [population/model]. To test this, we conducted a [design, e.g. randomized controlled trial/cohort study/lab experiment] in which we [briefly describe main approach]. This study aimed to determine [primary objective] and to explore [secondary objectives]." (Typically ~3–5 paragraphs, setting up rationale).

- **Methods (Example):** "We performed a [study design, e.g. double-blind RCT/retrospective cohort/animal study] at [institution] from [dates]. *Participants:* We enrolled N patients aged X–Y years with [condition], excluding those with [exclusion]. *Intervention/Exposure:* [Describe what was done or measured – e.g., "Participants were randomized to [treatment] or [placebo] for 12 weeks."]. *Data Collection:* Clinical data and [specimen] were collected at baseline and [follow-up]. *Outcomes:* The primary outcome was [describe outcome and how measured]. Secondary outcomes included [list]. *Statistical Analysis:* We calculated that N = [sample size] would detect a difference of [effect size] with 80% power (α=0.05). Continuous variables are presented as mean±SD (or median [IQR]) and compared with [t-test or Mann-Whitney]. Categorical variables were compared by χ^2. We used [software, version] for analysis. A *p*-value <0.05 was considered significant. *Ethics:* The study was approved by [IRB name, number], and all participants gave written informed consent." [12] [11]

- **Results (Example):** "Between [dates], we screened X patients and enrolled N (Figure 1). Baseline characteristics were similar between groups (Table 1). The mean age was … (SD), % female, etc. *Primary Outcome:* At 12 weeks, [treatment] significantly [increased/decreased] the primary endpoint compared to control (mean difference, 95% CI; *p*=…). *Secondary Outcomes:* [Summarize key secondary results, e.g. "Symptom score improved by X in the treatment group vs Y in placebo (p=…)"]. *Subgroup Analyses:* Effects were consistent across subgroups (supplemental data). *Adverse Events:* [If applicable] AEs occurred in X% of patients in the treatment group vs Y% in control (no serious AEs)." (Reference figures/tables in text, e.g. *"(Table 1; Figure 2)"*.)

- **Discussion (Example):** "In this [design] study, we found that [summary of main findings in 1–2 sentences], consistent with [previous studies or theory]. This suggests that [interpretation of mechanism]. Notably, we observed that [any unexpected or secondary finding]. Our findings support the hypothesis that [implied conclusion]. *Strengths:* include [novelty, rigorous design, sample size]. *Limitations:* This study has limitations: e.g., [single-center, observational design, small N, short follow-up, etc.]. These may affect [generalizability, power]. *Comparison:* Previous research by [cite] found [similar/different results], which [agree/disagree]. The difference may be due to [reasons]. *Implications:* Clinically, these results imply [how practice or understanding may change]. Mechanistically, [what new insight]. *Future directions:* Further trials with larger sample or longer follow-up are needed. In conclusion, [intervention] [does what] in [disease/model], indicating [final take-home message]."

- **"What You Need to Know" (Example):**

- *Background and Context:* "[Concise 25–30 words on disease context]."

- *New Findings:* "In patients with [condition], [intervention] significantly [effect] vs [control] in X-week trial."
- *Limitations:* "Study was limited by [small sample, short follow-up, etc.] and was conducted at a single center."
- *Clinical Research Relevance:* "This highlights [clinical implication, e.g. 'the potential of intervention X to reduce disease symptoms'] and guides future trials."

- *Basic Research Relevance:* "Our findings suggest [biological insight, e.g. 'a novel pathway for disease modulation'], providing a rationale for mechanistic studies."

- **Lay Summary (Example):** "We found that [in simple terms, e.g. 'a new diet reduced belly fat in people with IBS'] over [timeframe]. This suggests [real-world implication, e.g. 'diet changes may improve IBS symptoms']."

- **Acknowledgments (Example):** "We thank [Name] for technical support and [Name] for helpful discussions. This work was funded by [NIH grant number, Foundation, etc.]. Author J.D. had full access to all data and is responsible for its integrity. No additional medical writing assistance was used."

- **References (Example Style):** (Listed at end; see *Checklist*.) For example:

- Jones RS, et al. Gastroenterology 2011;140:123-130.
- Miller AB, Smith CD, et al. Gastroenterology 2026;170:101-110.

- Chen EF, et al. Gastroenterology 2025;169:45-53.
(Only the first author is named if >3 authors; shared first authors in bold.) 15

- **Figure Legends (Example):**

- *Figure 1:* CONSORT flow diagram of patient enrollment (n screened, excluded, randomized, analyzed).
- *Figure 2:* Primary outcome results. Bar graph showing mean ± SEM of [outcome] for each group (***p***<0.001 vs control).
- **Tables (Example):**
- *Table 1:* Baseline characteristics of participants (e.g., age, sex, BMI, lab values) by group. (Use means±SD or n (%) as appropriate.)

(Mock data for Table 1 shown below for illustration.)

Characteristic	Treatment (n=50)	Control (n=50)	*p*-value
Age, mean ± SD (yrs)	50.2 ± 10.3	49.8 ± 11.0	0.85
Female, n (%)	30 (60%)	28 (56%)	0.68
BMI, mean ± SD	27.5 ± 3.2	27.8 ± 3.5	0.70
Outcome Y, mean ± SD	12.3 ± 2.5	10.1 ± 2.8	<0.001

Table 1. **Baseline characteristics.** BMI: body mass index; SD: standard deviation. The treatment group had a higher primary outcome *Y* at baseline (p<0.001).

(The above table is a visual mockup; actual values will depend on real data.)

Submission Checklist

- **Manuscript Type:** Confirm submission as an *Original Article*. (Alternatively, if a brief report: see *Gastroenterology* guidelines for *Research Letters* [18].)
- **Word Count:** ≤7000 words total (includes text, references, table/figure legends) [19]. ~1000 words should be allocated to *Materials and Methods*.
- **Abstract:** ≤260 words, structured into Background/Aims; Methods; Results; Conclusions; plus 4–5 keywords [6]. Include trial registration number at end of abstract if a clinical trial [20].
- **Title Page:** Include title, full author list, affiliations, corresponding author information, conflict of interest disclosures for all authors, author contributions, ethical approvals, and data availability statement [1] [12]. For clinical trials: include *Clinical Trial Registration* (registry and ID) on title page.
- **Authorship:** All listed authors must meet ICMJE authorship criteria [3]. Show shared first-authorship by bolding names. Limit authors as per article type (Original Articles have no fixed max; Research Letters max 6 [21]).
- **Ethics:** For human studies, include IRB approval and informed consent statement; for animal studies, include IACUC approval and adhere to ARRIVE guidelines [12] [13]. Mention these approvals in *Methods*.
- **Reporting Guidelines:** Attach relevant checklists: **CONSORT** (RCT) [22], **STROBE** (observational) [23], **PRISMA** (systematic review) [5], **ARRIVE** (animal) [13], etc. Include flow diagrams (CONSORT/STARD) as figures if applicable [24].
- **Statistical Reporting:** Describe methods in detail (sample size calculation, handling of missing data, software/version) [11]. Report exact *p*-values, confidence intervals, and effect sizes (e.g. mean±SD, OR (95% CI)). Avoid ambiguous phrases (e.g. “not significant”).
- **Results and Data:** Present clear numeric results. For clinical trials, include baseline comparability and a CONSORT flowchart (as Figure) [22]. For observational studies, describe cohort/sample selection fully. Cite EQUATOR website for additional guidelines [25].
- **Figures/Tables:** Maximum 7 total (figures and tables combined) [26]. Figures should be separate high-resolution files (JPEG/TIFF/EPS/PDF, 300 dpi) [26]. Label figure panels clearly (A, B, etc.) and include legends in the main text. Tables must fit page width and not duplicate text results.
- **Supplementary Material:** Label any additional figures/tables as “Supplementary” in legend. Limit use of supplementary data to needed detail (e.g. extended methods, extra results). The main text should stand alone.
- **Conflict of Interest:** All authors must declare conflicts on the title page [27]. Use the journal’s conflict disclosure form.
- **Data Sharing:** For trials, include a *Data Transparency Statement* on the title page specifying if deidentified participant data will be shared, what, where, and for how long [17].
- **Copyright:** *Gastroenterology* requires authors to sign Elsevier’s license (no prior publication, no copyright infringement). Check *Gastroenterology*’s policies on image permissions and patient privacy (no PHI) [28].
- **Language:** The manuscript should be in clear, professional English. Non-native speakers should consider language editing. Maintain scientific tone suitable for a broad GI audience.
- **Timeline:** (Optional) Indicate planned revision or embargo dates in cover letter if needed.

Failure to follow these requirements (format, ethics, word counts, etc.) may delay review or cause rejection [2] [12].

References (Journal Style)

Format examples (Gastroenterology style, no article titles, first author only listed if >3 authors):

- Doe J, et al. Gastroenterology 2026;170:101-110.
- Smith AB, Johnson CD, et al. Gastroenterology 2025;169:45-53.
- Lee EF, et al. Gastroenterology 2024;168:200-210.

(Note: Shared first authors should be in **bold** in the list, with the footnote "*Author names in bold designate shared co-first authorship*." [15] . Data sets cited as: `[dataset] LastName, Initials. *Dataset Title*. Repository. Year. DOI`.)

Figures and Tables (Suggestions)

- **Figure 1:** *Patient Enrollment Flowchart.* A CONSORT-style diagram showing numbers screened, excluded (with reasons), randomized, and analyzed in each arm. (Diagram created per CONSORT template [22] .)
- **Figure 2:** *Primary Outcome by Group.* Bar or line graph of the main quantitative result. *Mockup:* Bar chart showing mean±SEM of outcome *Y* for Treatment vs Control (**p<0.01**, *n*=50 each). (Caption should describe the test used and significance.)
- **Figure 3:** *Representative Data.* Example: A microscopy image or physiological tracing relevant to the study. (Ensure no patient identifiers.)
- **Table 1:** *Baseline Characteristics.* Demographics and key clinical/lab values for each group (as above).
- **Table 2:** *Main Results.* E.g., primary and secondary outcomes with effect estimates (mean ± SD or % with 95% CI and p-values).

Figure/Table layout mockups: (Illustrations of possible layouts; actual figures will depend on data.)

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[Example table layout as above showing baseline data; can also be drawn in
software with proper formatting.]
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Figure: Suggested manuscript workflow (conceptual flowchart).

Timeline and Task Breakdown

A sample Gantt-chart timeline for writing and submission (assuming a 6-month schedule) is shown below. Tasks can overlap. Adjust durations based on project specifics.

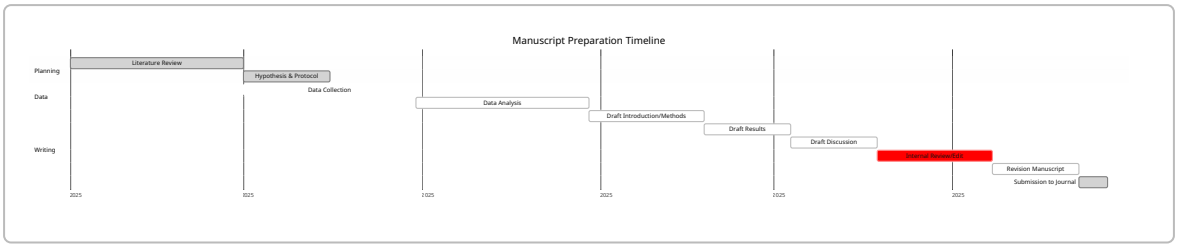


Figure: Proposed timeline (months) and tasks leading to submission.

Tasks by week:
1. **Weeks 1–4:** Finalize study protocol, get approvals, complete data collection.
2. **Weeks 5–6:** Analyze data; generate tables/figures.
3. **Weeks 7–8:** Write *Methods* and *Results* sections; create initial figures/tables.
4. **Weeks 9–10:** Write *Introduction* and *Discussion*; draft abstract.
5. **Weeks 11–12:** Internal review (co-authors and statisticians) and edit.
6. **Weeks 13–14:** Revise manuscript; finalize *What You Need to Know* and *Lay Summary*.
7. **Week 15:** Format references, prepare cover letter, and submit.

Use collaborative writing (version control) and allow buffer time for co-author review. A table summarizing responsibilities may help (e.g., “Primary analyst: X; Writing lead: Y; Literature search: Z”).

References

(Guidance and requirements cited from: Gastroenterology Author Guidelines [1] [2] [4] [5]; ICMJE Recommendations [3]; EQUATOR Network (CONSORT/STROBE/PRISMA) [4] [5]; and relevant literature.)

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[1] [2] [4] [5] [6] [7] [8] [9] [10] [12] [13] [14] [15] [16] [17] [18] [19] [20] [21] [22] [23] [24] [25] [26] [27] [28] Guide for authors - Gastroenterology - ISSN 0016-5085 | ScienceDirect.com by Elsevier
https://www.sciencedirect.com/journal/gastroenterology/publish/guide-for-authors

[3] ICMJE | Recommendations | Defining the Role of Authors and Contributors
https://www.icmje.org/recommendations/browse/roles-and-responsibilities/defining-the-role-of-authors-and-contributors.html

[11] Guidelines for Statistical Reporting in Medical Journals - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC7642026/